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Editor Radiation Physics and Chemistry

Subject: Submission of manuscript for consideration in Radiation Physics and Chemistry

Dear Editor,

I am pleased to submit the manuscript “Memory-Efficient Geant4 Modelling Regimes for Nanocomposite Radiation Shielding: Effective Medium, Explicit Nanoparticle Geometry, and Hybrid Representative-Volume Validation” for consideration as an original research article in Radiation Physics and Chemistry.

The manuscript is well aligned with the journal’s readership because it addresses photon attenuation, shielding-material modelling, and Monte Carlo transport methodology in a directly usable form. The study benchmarks three Geant4 modelling regimes for an HDPE/Bi₂O₃ nanocomposite at matched composition and density: an effective-medium material model, an explicit G4PVParameterised representative-volume geometry, and an explicit G4MultiUnion representative-volume geometry.

The central result is that all three regimes agree within the pre-registered 3% acceptance window at 30-150 keV for the matched HDPE/Bi₂O₃ benchmark system. This supports the effective-medium approximation for macroscopic attenuation when the photon mean free path greatly exceeds the nanoparticle radius. The study also contributes a practical computational result: G4PVParameterised remains tractable up to a 1 μ m representative volume containing 2,888 explicit nanoparticles, whereas G4MultiUnion becomes impractical beyond about 45 spheres on 8 GB workstation-class hardware because of voxelizer growth.

The manuscript therefore offers a concrete workflow recommendation for shielding studies: use the effective-medium model as the production default, use explicit parameterised representative volumes as a validation geometry when explicit structure must be checked, and reserve G4MultiUnion for small-cluster stress tests rather than dense large-RVE production work. The novelty is methodological rather than kernel-level, namely a controlled cross-regime shielding benchmark and representative-volume validation workflow implemented in a reproducible open benchmark harness.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I confirm that I am the sole author and that I approve the submitted version. There are no conflicts of interest to declare, and no external funding supported this work.

Sincerely,

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